
Exercice 1: gradient Lipschitz and (strongly) smooth functions

Let $L > 0$ fixed. In this exercise, a function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be :

(P1) **gradient Lipschitz** (with constant L) if and only if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all $x, y \in \mathbb{R}^n$.

(P2) **strongly smooth** (or L-smooth) if and only if $\frac{L}{2}\|.\|^2 - f(.)$ is convex.

For convex functions, **(P1)** and **(P2)** are equivalent, which explains why both notions are often considered equivalent. In the general non convex case, we have only **(P1)** $\Rightarrow$ **(P2)**. Proofs are detailed in this exercise.

1. Give an example of a non convex function that is : L-smooth but non-differentiable / differentiable but not gradient Lipschitz.

   In the following questions, we consider only *differentiable* functions.

2. Prove that a function g is convex if and only if its gradient is monotone, that is :

$$\langle \nabla g(x) - \nabla g(y), x - y \rangle \geq 0 \qquad (\forall x, y \in \mathbb{R}^n)\,.$$

3. Prove that gradient Lipschitz implies strongly smooth, that is : **(P1)** $\Rightarrow$ **(P2)**.

4. Prove that f is L-smooth if and only if :

(P2') For all x, y $f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2}\|x - y\|^2$.

5. We want to prove now that, for convex functions, L-smooth is equivalent to gradient Lipschitz. The suggested proof goes through the following properties :

(P3) $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2$ for all $x, y \in \mathbb{R}^n$.

(P4) $1/L$**-cocoercivity** of gradient : $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2$ for all $x, y \in \mathbb{R}^n$.

   (a) Show that **(P3)** $\Rightarrow$ **(P4)**.

   (b) Show that $1/L$-cocoercivity implies gradient L-Lipschitz, that is **(P4)** $\Rightarrow$ **(P1)**.

   (c) Finally, using convexity of the function, prove **(P2)** $\Rightarrow$ **(P3)** (Hint : introduce $\phi_x(z) = f(z) - \langle \nabla f(x), z \rangle$ and note that it is L-smooth).

   The latter result is often known as the **Baillon-Haddad** theorem, which states that a convex gradient Lipschitz function is $1/L$ cocoercive.

Réponses exercice: 1

1. $f(x) = -|x|$ satisfies **(P2)** but is not differentiable.
   $f(x) = -x^4$ satisfies **(P2)** but its gradient (derivative) $f'(x) = -4x^3$ is not Lipschitz on $\mathbb{R}$.

2. A differentiable function g is convex if and only if for all x, y :

$$g(x) + \langle \nabla g(x), y - x \rangle \leq g(y) \text{ and also by exchanging } x \text{ and } y$$
$$g(y) + \langle \nabla g(y), x - y \rangle \leq g(x)\,.$$

   Summing the two inequalities, g is convex if and only if $\langle \nabla g(x) - \nabla g(y), x - y \rangle \geq 0$ for all x, y.

3. Assume **(P1)**. By the Cauchy-Schwarz inequality, we obtain :

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \leq \|\nabla f(x) - \nabla f(y)\|\,\|x - y\| \leq L\|x - y\|^2$$

   and hence :

$$0 \leq \langle Lx - \nabla f(x) - Ly + \nabla f(y), x - y \rangle\,.$$

   This shows that the gradient of $g(.) = \frac{L}{2}\|.\|^2 - f(.)$ is monotone and hence f is L-smooth by **(P2)**.

4. According to **(P2)**, f is L-smooth if and only if $g(.) = \frac{L}{2}\|.\|^2 - f(.)$ is convex and this reads :

$$g(y) + \langle \nabla g(y), x - y \rangle \leq g(x) \qquad (\forall x, y)$$
$$\text{equivalently : } \frac{L}{2}\|y\|^2 - f(y) + \langle Ly - \nabla f(y), x - y \rangle \leq \frac{L}{2}\|x\|^2 - f(x) \qquad (\forall x, y)\,.$$

   Rearranging, we obtain **(P2')**.

5. (a) Assume **(P3)** and hence for all x, y we have, where both variables have been exchanged in the second line :

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2$$
$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{1}{2L}\|\nabla f(x) - \nabla f(y)\|^2$$

and hence, summing both inequalities, we get **(P4)** :

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2 .$$

(b) Suppose $\nabla f(.)$ is $1/L$-cocoercive and use the Cauchy-Schwarz inequality :

$$\frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2 \leq \langle \nabla f(x) - \nabla f(y), x - y \rangle \leq \|\nabla f(x) - \nabla f(y)\|\|x - y\| .$$

After simplification, we obtain that $\nabla f(.)$ is L-Lipschitz.

(c) Introduce $\phi_x(z) = f(z) - \langle \nabla f(x), z \rangle$. It is a convex function, with minimum reached at $z^\star = x$. In addition, $\phi_x(.)$ satisfies L-smoothness, which gives :

$$\frac{L}{2}\|z\|^2 - \phi_x(z) \geq \frac{L}{2}\|y\|^2 - \phi_x(y) + \langle Ly - \nabla\phi_x(y), z - y \rangle$$
$$\phi_x(z) \leq \phi_x(y) + \langle \nabla\phi_x(y), z - y \rangle + \frac{L}{2}\left[\|z\|^2 + \|y\|^2 - 2\langle y, z \rangle\right]$$
$$\phi_x(z) \leq \phi_x(y) + \langle \nabla\phi_x(y), z - y \rangle + \frac{L}{2}\|z - y\|^2$$

Minimize now above on the left hand side , and then on the right hand side with respect to z. We get :

$$\phi_x(x) \leq \phi_x(y) - \frac{1}{2L}\|\nabla\phi_x(y)\|^2$$
$$f(x) - \langle \nabla f(x), x \rangle \leq f(y) - \langle \nabla f(x), y \rangle - \frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2$$
$$\frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2 \leq f(y) - f(x) - \langle \nabla f(x), y - x \rangle .$$

This is **(P3)** and concludes the proof.

Exercice 2: coercivity, strong convexity, optimal step gradient descent

1. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and $\lim_{\|x\| \to \infty} f(x) = +\infty$. Show that there exist a point $x^\star$ such that $f(x^\star)$ is minimum.
2. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable and satisfies μ-strong convexity, that is, for a given $\mu > 0$:

$$f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|^2 \leq f(y) \qquad (\forall x, y)$$

Show that there exist a unique minimizer $x^\star$ and that $\nabla f(x) = 0 \Rightarrow x = x^\star$.

3. Given an initialization point $x_0 \in \mathbb{R}^n$, consider the gradient descent algorithm with optimal step :

$$\forall k \in \mathbb{N} : \quad x_{k+1} = x_k - \gamma_k \nabla f(x_k) \text{ with } \gamma_k = \operatorname{Arg\,min}_{t \in \mathbb{R}} f(x_k - t\nabla f(x_k)) .$$

Prove that $x_k \xrightarrow[k \to +\infty]{} x^\star$.

Réponses exercice: 2

1. Let $A > 0$ such that $\|x\| > A \Rightarrow f(x) > f(0)$. The closed ball $\{x \in \mathbb{R}^n \mid \|x\| \leq A\}$ is compact and by continuity f reaches in the ball a minimum at a point $x^\star$. The latter is a global minimum of f because $f(x) \geq f(x^\star)$ for any x in the ball and $f(x) > f(0) \geq f(x^\star)$ for any x outside the ball.
2. By the inequality of strong convexity, $\lim_{\|x\| \to \infty} f(x) = +\infty$ and by the previous question there exists a point $x^\star$ where f reaches its minimum.
Suppose $\nabla f(x) = 0$. Then by the inequality for strong convexity first, and by the fact that f reaches its minimum at $x^\star$:

$$f(x) + \frac{\mu}{2}\|x^\star - x\|^2 \leq f(x^\star) \leq f(x) .$$

As a consequence, the minimum is unique and $x = x^\star$ for any x such that $\nabla f(x) = 0$.

3. Note first that the sequence $(x_k)_{k\in\mathbb{N}}$ is well defined. If $\nabla f(x_k)=0$, then $x_{k+1}=x_k$; otherwise use strong convexity and $f(x_k - t\nabla f(x_k)) \xrightarrow[t\to+\infty]{} +\infty$ to see that γ_k is well defined and positive.

   In addition, γ_k is such that

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x_k - t\nabla f(x_k))\Big|_{\gamma_k} = -\nabla f(x_k)^\top \nabla f(x_k) - \gamma_k \nabla f(x_k)) = 0\,.$$

   Therefore $\nabla f(x_k)^\top \nabla f(x_{k+1}) = 0$.
   By construction $(f(x_k))_{k\in\mathbb{N}}$ is decreasing and lower bounded by $f(x^\star)$. $\lim_{k\to\infty} f(x_k)$ hence exists and $(f(x_k))_{k\in\mathbb{N}}$ is bounded in particular. By coercivity, $(x_k)_{k\in\mathbb{N}}$ is hence also bounded. We write $x_{\lim}$ for a limit point of the sequence obtained as the limit of a subsequence $(x_{s_k})_k$. Then :

$$\frac{\mu}{2}\|x_{s_k} - x_{s_k+1}\|^2 \le f(x_{s_k}) - f(x_{s_k+1}) \to 0$$

   $(x_{s_k+1})_k$ hence also converges to $x_{\lim}$. Finally, for all k, $0 = \nabla f(x_{s_k})^\top \nabla f(x_{s_k+1}) \to \|\nabla f(x_{\lim})\|^2$. It follows $\nabla f(x_{\lim}) = 0$ and hence $x_{\lim} = x^\star$.

Exercice 3: strongly smooth, strongly convex function, fixed step gradient descent

We consider in this exercise a convex function $f : \mathbb{R}^n \to \mathbb{R}$.

1. We assume that f is μ-strongly convex, that is $f(.) - \frac{\mu}{2}\|.\|^2$ is convex. Prove :
   (a) $f(y) \ge f(x) + \langle \nabla f(x), y-x\rangle + \frac{\mu}{2}\|y-x\|^2$ for all x, y.
   (b) f admits a unique minimizer $x^\star$ which is given by $\nabla f(x^\star) = 0$. We note below $f^\star = f(x^\star)$.
   (c) $\frac{2}{\mu}(f(x) - f^\star) \ge \|x - x^\star\|^2$ for all x.
   (d) $\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f^\star))$ (**(PL)** Polyak-Lojasiewicz condition)
2. We assume in addition that f has L-Lipschitz gradient and we recall that the following holds for all x, y (see other exercise) :

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\| \qquad \text{(gradient Lipschitz)}$$

$$f(y) \le f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2 \qquad (L\text{-smoothness})$$

$$\frac{1}{L}\|\nabla f(x) - \nabla f(y)\|^2 \le \langle \nabla f(x) - \nabla f(y), x-y\rangle \qquad \text{(cocoercivity of gradient)}$$

   Consider the gradient algorithm with fixed step $\gamma > 0$ defined by $x_{k+1} = x_k - \gamma\nabla f(x_k)$ for all $k \in \mathbb{N}$.
   (a) Prove that $f(x_{k+1}) \le f(x_k) - \gamma\left(1 - \frac{\gamma L}{2}\right)\|\nabla f(x_k)\|^2$ (**descent lemma**).
   (b) Prove that $f(x_{k+1}) - f^\star \le \left(1 - 2\gamma\mu\left(1 - \frac{\gamma L}{2}\right)\right)(f(x_k) - f^\star)$.
   (c) For $\gamma = 1/L$, prove that : $f(x_{k+1}) - f^\star \le (1 - \mu/L)(f(x_k) - f^\star)$ and $f(x_k) - f^\star \le \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^\star)$.
   (d) <u>Remark :</u> (nothing to prove) note that convergence can be deduced on $(x_k)_{k\in\mathbb{N}}$ by the arguments above. Other results can be obtained by showing that the gradient operator $T_\gamma(x) = x - \gamma\nabla f(x)$ is a contraction.

Réponses exercice: 3

1. (a) Convexity of $h(x) = f(x) - \frac{\mu}{2}\|x\|^2$ is equivalent to :

$$f(y) - \frac{\mu}{2}\|y\|^2 \ge f(x) - \frac{\mu}{2}\|x\|^2 + \langle \nabla f(x) - \mu x, y - x\rangle \qquad (\forall x, y)$$

   This proves the result.
   (b) Use previous question (see other exercise).
   (c) Use previous two questions (with $y = x^\star$).
   (d) Minimize right hand side of inequality proven in first question with respect to y and get :

$$\begin{aligned} f^\star \ge f(y) &\ge f(x) + \langle \nabla f(x), y-x\rangle + \frac{\mu}{2}\|y-x\|^2 \\ &\ge f(x) - \frac{1}{\mu}\langle \nabla f(x), \nabla f(x)\rangle + \frac{\mu}{2}\|\frac{1}{\mu}\nabla f(x)\|^2 \qquad \text{(obtained for } y = x - \frac{1}{\mu}\nabla f(x)) \\ &\ge f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2\,. \end{aligned}$$

   This is (PL) condition.

2. (a) From the L-smoothness inequality with $y = x_{k+1}$ and $x * x_k$:

$$f(x_{k+1}) \leq f(x_k) + \langle \nabla f(x_k), -\gamma \nabla f((x_k)) \rangle + \frac{L}{2} \| - \gamma \nabla f(x_k)\|$$
$$\leq f(x_k) - \gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla f(x_k)\|^2 .$$

This is the sough descent lemma.

(b) Use the descent lemma and the (PL) condition :

(c) Replace $\gamma = 1/L$ in the previous inequality.

(d) Remark : that's not the end of the story indeed !

Exercice 4: fixed step gradient descent for strongly smooth function

We consider a function f convex, gradient Lipschitz (constant L, see other exercise for implications of the gradient Lipschitz assumption) and we assume that f admits a minimizer $x^\star$ that is non necessarily unique and satisfies by $\nabla f(x^\star) = 0$. We denote $f^\star = f(x^\star)$ the minimum value of f.

We derive here some convergence results for the gradient algorithm with fixed step $\gamma > 0$ defined by $x_{k+1} = x_k - \gamma \nabla f(x_k)$ for all $k \in \mathbb{N}$.

1. Prove that $f(x_{k+1}) \leq f(x_k) - \gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla f(x_k)\|^2$ (**descent lemma**).
2. Show that $(f_{x_k})_{k\in\mathbb{N}}$ is a decreasing sequence for a range of γ to be specified and prove that $\lim_{k\to+\infty} \|\nabla f(x_k)\| = 0$.
3. Prove that for any minimizer $x^\star$, we have : $\|x_{k+1} - x^\star\|^2 \leq \|x_k - x^\star\|^2 - \gamma(\frac{2}{L} - \gamma)\|\nabla f(x_k)\|^2$ (Hint : use the cocoercivity of the gradient). Deduce successively :
   — $(\|x_k - x^\star\|)_{k\in\mathbb{N}}$ is decreasing for any $x^\star$ (this is called a Féjer-monotone sequence with respect to the set of minimizers of f).
   — $(x_k)_{k\in\mathbb{N}}$ is a bounded sequence.
   — $(x_k)_{k\in\mathbb{N}}$ converges to a minimizer of f.
4. Prove that :

$$f(x_N) - f^\star \leq \frac{\|x_0 - x^\star\|^2}{2\gamma N} + \frac{f(x_0 - f^\star)}{2N(1 - L\gamma/2)}$$

Remark : depending on the technique/assumptions, other constants may be found but in any case, a $O(1/N)$ convergence rate is obtained.

Réponses exercice: 4

1. From the L-smoothness inequality with $y = x_{k+1}$ and $x * x_k$:

$$f(x_{k+1}) \leq f(x_k) + \langle \nabla f(x_k), -\gamma \nabla f((x_k)) \rangle + \frac{L}{2} \| - \gamma \nabla f(x_k)\|$$
$$\leq f(x_k) - \gamma \left(1 - \frac{\gamma L}{2}\right) \|\nabla f(x_k)\|^2 .$$

This is the sough descent lemma.

2. For $\gamma \in]0, 2/L[$, we see from the descent lemma that $(f_{x_k})_{k\in\mathbb{N}}$ is a decreasing sequence, hence admits a limit because it is lower-bounded by $f^\star$. By summing the descent lemma inequality for $k = 0$ to $k = N-1$, we get :

$$\sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 \leq \frac{f(x_0) - f(x_N)}{\gamma(1 - \gamma L/2)} \text{ and hence : } \sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \leq \frac{f(x_0) - f^\star}{\gamma(1 - \gamma L/2)} .$$

The above sequence is summable and, which proves $\lim_{k\to+\infty} \|\nabla f(x_k)\|^2 = 0$.

3. For any minimizer $x^\star$:

$$\|x_{k+1} - x^\star\|^2 = \|x_k - \gamma \nabla f(x_k) - x^\star\|^2$$
$$= |x_k - x^\star\|^2 - 2\gamma \langle \nabla f(x_k), x_k - x^\star \rangle + \gamma^2 \|\nabla f(x_k)\|^2$$

Since $\nabla f(x^\star) = 0$, by cocoercivity of the gradient $\langle \nabla f(x_k), x_k - x^\star \rangle \geq \frac{1}{L}\|\nabla f(x_k)\|^2$. Hence :

$$\|x_{k+1} - x^\star\|^2 \leq \|x_k - x^\star\|^2 - \gamma(\frac{2}{L} - \gamma)\|\nabla f(x_k)\|^2$$

The following follows :

— $(\|x_k - x^\star\|)_{k\in\mathbb{N}}$ is decreasing for any $x^\star$
— $(x_k)_{k\in\mathbb{N}}$ is a bounded sequence (take any $x^\star$). As a consequence any accumulation point is a minimizer because $\lim_{k\to\infty} \|\nabla f(x_k)\| = 0$.
— $(x_k)_{k\in\mathbb{N}}$ converges to a minimizer of f.

4. By convexity, $f(x_k) + \langle \nabla f(x_k), x^\star - x_k\rangle \leq f^\star$ and hence

$$2\gamma(f(x_k) - f^\star) \leq 2\langle \gamma\nabla f(x_k), x_k - x^\star\rangle = \|x_k - x^\star\|^2 - \|x_{k+1} - x^\star\|^2 + \|\gamma\nabla f(x_k)\|^2 .$$

Summing now, we get :

$$2\gamma \sum_{k=0}^{N-1} (f(x_k) - f^\star) \leq \|x_0 - x^\star\|^2 - \|x_N - x^\star\|^2 + \gamma^2 \sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 .$$

This can be simplified because $(f(x_k))_k$ decreases, and the formerly obtained upper-bound can also be introduced :

$$2\gamma N(f(x_N) - f^\star) \leq \|x_0 - x^\star\|^2 + \gamma^2 \sum_{k=0}^{N-1} \|\nabla f(x_k)\|^2 \leq \|x_0 - x^\star\|^2 + \gamma^2 \frac{f(x_0) - f^\star}{\gamma(1 - \gamma L/2)} .$$

Dividing by $2\gamma N$, we obtain the result.